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1 Query Plans

1.1 Namespace log N1

- Create a namespace:

```
kubectl create namespace rising
```

- Get the namespace:

```
kubectl get namespaces
```

- Describe the namespace:

```
kubectl describe namespace rising
```

- Delete the namespace:

```
kubectl delete namespace rising
```

1.2 Pod log

- Create a pod:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: rising-pod
    namespace: rising
  spec:
    containers:
    - name: first-container
      image: nginx
    - name: second-container
      image: alpine
      command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 36000"]
EOF
```

- Inspect the Pod:

```
kubectl -n rising describe pod rising-pod
```

- Patch the Pod and try to add a third container, this will fail:

```
kubectl -n rising patch pod rising-pod --type='json' -p='[{"op": "add", "path":
↪ "/spec/containers/1", "value": {"name": "third-container", "image": "nginx"}}]'
```

- Get the pod log:

```
kubectl -n rising logs rising-pod -c second-container
```

- Execute commands in the second containers, the first will complain about the shell, the second will work:

```
kubectl -n rising exec -t rising-pod -c second-container -- /bin/bash -c "echo Hello again,  
↵ Kubernetes!"  
kubectl -n rising exec -t rising-pod -c second-container -- /bin/sh -c "echo Hello again,  
↵ Kubernetes!"
```

- Delete the pod:

```
kubectl -n rising delete pod rising-pod
```

1.3 Pod log P2 - Image download

1.4 ConfigMap C1 - Create a ConfigMap and use it in a Pod, then delete it

1.5 Secret S1 - Create a Secret and use it in a Pod, then delete it